

ECG Simulator Documentation
Part 4: GUI Reference & Parameter Reference

Comprehensive Technical Documentation

June 2026

Contents

1 Main Window Architecture

1.1 Overview

The PyQt6-based GUI provides real-time visualization of 12-lead ECG with interactive parameter control. The main window is organized into three primary regions:

- **Control Panel** (left): Heart rate, engine selection, beat controls
- **ECG Display** (center): Real-time scrolling 12-lead ECG trace
- **Parameter Panel** (right): 50+ physiological parameter controls

1.2 Control Panel Layout

1.2.1 Heart Rate (HR) Control

- **Widget Type:** QSpinBox
- **Range:** 30–220 bpm
- **Default:** 70 bpm
- **Step:** 1 bpm
- **Effect:** Directly sets SA node cycle length via:

$$\text{cycle_length} = \frac{60}{HR} \quad (1)$$

- **Clinical Range:** 60–100 bpm (normal sinus rhythm)
- **Bradycardia:** $\leq$ 60 bpm (typically 40–60 in athletes)
- **Tachycardia:** $\geq$ 100 bpm (up to 200+ during exercise)

1.2.2 Simulation Engine Selector

- **Widget Type:** QComboBox
- **Options:**
 - **graph** (Phase 1/2: BFS-based, $\leq$ 1000 Hz, deterministic)
 - **cable** (Phase 3: FHN cable, 2.2 $\times$ real-time, physiologically accurate)
- **Default:** graph
- **Effect:** Switches underlying simulation engine
- **Behavior:** Resets current beat on switch

1.2.3 Beat Control Buttons

- **Start Button:** Begin continuous rhythm generation
 - Starts simulation timer (33 ms per frame @ 30 FPS)
 - Disables beat controls during active simulation
- **Stop Button:** Pause simulation (preserves current ECG trace)
 - Allows parameter adjustment without clearing ECG
- **Reset Button:** Clear ECG trace and restart
 - Clears all accumulated beats from display
 - Resets internal beat counters

2 ECG Display Widget

2.1 Real-Time 12-Lead Display

- **Library:** pyqtgraph (for real-time plotting performance)
- **Update Rate:** 30 FPS
- **Scroll Behavior:** Left-to-right, continuous scroll
- **Time Window:** 10 seconds visible (adjustable)
- **Leads Displayed:** All 12 ECG leads (6 frontal + 6 precordial)

2.2 12-Lead Organization

2.2.1 Frontal Leads (rows 1–2)

Lead	Axis	Clinical Use
I	0° (left)	Left-right forces
II	60° (inferolateral)	Inferior/lateral forces
III	120° (inferior)	Inferior forces
aVR	-150° (right)	Right atrium/ventricle
aVL	-30° (left)	Left ventricle
aVF	90° (inferior)	Inferior wall

2.2.2 Precordial Leads (rows 3–4)

Lead	Anatomical Position	Chamber
V1	4th ICS, right sternal border	RV outflow
V2	4th ICS, left sternal border	LV outflow
V3	Midway V2–V4	LV septum
V4	5th ICS, midclavicular	LV apex
V5	5th ICS, anterior axillary	LV lateral
V6	5th ICS, midaxillary	LV lateral

3 Parameter Panel Organization

3.1 Structure

The parameter panel is organized into 12 collapsible groups:

1. SA Node Parameters
2. AV Node Parameters
3. His Bundle & Purkinje
4. Atrial Conduction
5. Ventricular Conduction
6. Ventricular Refractory Periods
7. Escape Pacemaker
8. Atrial Fibrillation
9. Ectopic Focus
10. ECG Scaling
11. Plugin Controls
12. Advanced/Debug

4 Complete Parameter Reference (50+ Parameters)

4.1 Group 1: SA Node Parameters

Parameter	Description	Range	Unit
cycle_length	Interval between SA fires	0.3–2.0	s
funny_current	HCN4 channel conductance	0.0–1.0	g/g_max
autonomic_tone	Sympathetic/parasympathetic balance	-1.0 to +1.0	normalized

4.1.1 Usage Notes

- **cycle_length**: Typically set via HR spinner; direct control available
- **funny_current**: Alters SA pacemaker rate
 - 0.0 = No spontaneous depolarization (requires external stimulus)
 - 1.0 = Maximum rate (200 bpm)
- **autonomic_tone**: Modulates HR via sympathetic/parasympathetic effects
 - -1.0 = Maximum parasympathetic (vagal): 40 bpm
 - 0.0 = Balanced (normal): 70 bpm
 - +1.0 = Maximum sympathetic: 120 bpm

4.2 Group 2: AV Node Parameters

Parameter	Description	Range	Unit
av_delay	Conduction time through AV node	0.05–0.30	s
av_refractory	AV node absolute refractory period	0.20–0.40	s
conductance	AV nodal conductance (0=block)	0.0–1.0	g/g_max

4.2.1 Clinical Significance

- **av_delay**: Normal = 100–200 ms
 - Controls PR interval via: $PR = P\text{-wave} + av_delay + QRS$
 - Prolonged PR (First-degree AV block) when ≥ 200 ms
- **conductance**: Controls AV block severity
 - 1.0 = Normal conduction (1:1 conduction ratio)
 - 0.5 = 2:1 AV block (every 2nd atrial beat blocked)
 - 0.0 = Complete AV block (III° AV block)

4.3 Group 3: His Bundle & Purkinje System

Parameter	Description	Range	Unit
his_delay	His bundle conduction time	0.01–0.05	s
lbb_conductance	Left bundle branch conductance	0.0–1.0	g/g_max
rbb_conductance	Right bundle branch conductance	0.0–1.0	g/g_max
lahb_conductance	Left ant. fascicle conductance	0.0–1.0	g/g_max
lphb_conductance	Left post. fascicle conductance	0.0–1.0	g/g_max

4.3.1 Bundle Branch Block Patterns

- **LBBB** (Left Bundle Branch Block):
 - Set `lbb_conductance` = 0.0, keep `rbb_conductance` = 1.0
 - ECG: Broad QRS (≥ 120 ms), M-shaped in V5–V6
- **RBBB** (Right Bundle Branch Block):
 - Set `rbb_conductance` = 0.0, keep `lbb_conductance` = 1.0
 - ECG: Broad QRS (≥ 120 ms), M-shaped in V1–V2
- **LAHB** (Left Anterior Hemiblock):
 - Set `lahb_conductance` = 0.0, keep `lphb_conductance` = 1.0
 - ECG: Left axis deviation, QRS ≥ 120 ms
- **LPHB** (Left Posterior Hemiblock):
 - Set `lphb_conductance` = 0.0, keep `lahb_conductance` = 1.0
 - ECG: Right axis deviation, QRS ≥ 120 ms

4.4 Group 4: Atrial Conduction

Parameter	Description	Range	Unit
<code>ra_conduction</code>	Right atrial conduction velocity	0.50–1.50	m/s
<code>la_conduction</code>	Left atrial conduction velocity	0.50–1.50	m/s
<code>atrial_refractory</code>	Atrial myocardium absolute refractory	0.20–0.35	s

4.5 Group 5: Ventricular Conduction

Parameter	Description	Range	Unit
<code>sept_conduction</code>	Septal conduction velocity	0.3–1.5	m/s
<code>lv_conduction</code>	LV free wall conduction velocity	0.30–1.0	m/s
<code>rv_conduction</code>	RV free wall conduction velocity	0.30–1.0	m/s

4.5.1 QRS Duration Formula

Total QRS duration depends on conduction velocities and geometric path lengths. For Phase 1/2 engine:

$$QRS_{\text{duration}} \approx \frac{\text{longest path}}{v_{\min}} \quad (2)$$

4.6 Group 6: Ventricular Refractory Periods

Parameter	Description	Range	Unit
<code>vent_abs_refr</code>	Absolute refractory period	0.25–0.45	s
<code>vent_rel_refr</code>	Relative refractory period	0.10–0.35	s

4.6.1 Physiological Basis

- **Absolute Refractory Period (ARP):** Voltage-gated Na^+ channels inactivated
 - No stimulus can trigger action potential
 - Normal duration: 200–300 ms
 - Set on Phase 1/2 engine only
- **Relative Refractory Period (RRP):** Partial recovery of Na^+ channel availability
 - Stimulus can trigger AP but with reduced amplitude
 - Conduction may be slower or blocked
 - Normal duration: 150–300 ms additional
- **Phase 3 Engine:** Refractory periods emerge naturally from FHN w variable
 - Recovery time determined by:

$$\tau_w = \frac{\tau_s}{b \cdot \varepsilon} = \frac{0.015}{0.8 \times 0.08} \approx 234 \text{ ms} \quad (3)$$

- No explicit refractory parameter needed

4.7 Group 7: Escape Pacemaker

Parameter	Description	Range	Unit
escape_enabled	Enable backup pacemaker	Yes/No	bool
escape_interval	Escape interval (interval without SA)	0.5–3.0	s
escape_origin	Ectopic site (AV/HIS/LV)	enum	-

4.7.1 Escape Pacemaker Mechanism

- Automatically fires if no atrial impulse received
- Clinical significance:
 - **Junctional escape** (origin=AV): Fires at 40–60 bpm, narrow QRS
 - **Ventricular escape** (origin=LV): Fires at 20–40 bpm, broad QRS
- escape_interval determines delay before firing:
 - If no stimulus received for $\geq$ escape_interval seconds, escape fires
 - Common clinical values: 1.5–3.0 seconds

4.8 Group 8: Atrial Fibrillation

Parameter	Description	Range	Unit
af_enabled	Enable atrial fibrillation	Yes/No	bool
af_rate	F-wave (fibrillatory wave) rate	300–700	bpm
af_rr_distribution	RR interval variation type	fixed/poisson/gaussian	enum
af_f_amplitude	F-wave amplitude	0.01–0.20	mV

4.8.1 AF ECG Characteristics

- **AF Rate:** Normal 300–600 bpm (fibrillatory waves from disorganized reentry)
 - Slower rates (300 bpm) suggest possible organized activity (flutter)
 - Faster rates (600+ bpm) suggest rapid fibrillatory activity
- **RR Distribution:**
 - fixed: Constant interval (used for testing)
 - poisson: Random exponential intervals (physiologic, irregular)
 - gaussian: Normal distribution around mean (moderate irregularity)
- **F-wave Amplitude:** Visible baseline oscillations
 - Coarse AF: ≥ 0.10 mV (more organized)
 - Fine AF: ≤ 0.05 mV (more disorganized)

4.9 Group 9: Ectopic Focus

Parameter	Description	Range	Unit
ectopic_enabled	Enable ectopic focus	Yes/No	bool
ectopic_coupling	Coupling interval (PVC timing)	0.40–0.80	s
ectopic_repetitive	Consecutive ectopics	Yes/No	bool
ectopic_origin	Ectopic site	LA/RA/LV/RV	enum

4.9.1 Premature Contraction Types

- **PAC (Premature Atrial Contraction):**
 - Set ectopic_origin = LA or RA
 - Normal coupling: 350–450 ms
 - Early coupling (≤ 350 ms) may block AV node $\rightarrow$ aberrant conduction
- **PVC (Premature Ventricular Contraction):**
 - Set ectopic_origin = LV or RV
 - Normal coupling: 400–600 ms
 - ECG: Broad QRS (≥ 120 ms), T-wave opposite normal polarity
 - Couplets/triplets possible if ectopic_repetitive = true

4.10 Group 10: ECG Scaling

Parameter	Description	Range	Unit
amplitude_scale	Global amplitude multiplier	0.1–5.0	unitless
baseline_shift	DC offset for visibility	-2.0–+2.0	mV

5 Advanced Controls & Debug Parameters

5.1 Group 11: Plugin Controls

- **Active Plugins:** Dropdown showing enabled pathologies
- **Plugin Stacking:** Select multiple pathologies simultaneously
- **Save/Load Presets:** Store/retrieve parameter combinations

5.2 Group 12: Advanced/Debug

Parameter	Description	Range	Unit
debug_mode	Enable diagnostic output	Yes/No	bool
show_activation_times	Print node activation times	Yes/No	bool
cable_internal_dt	Phase 3 integration step	0.0001–0.001	s

6 Parameter Adjustment Workflow

6.1 Typical Use Cases

6.1.1 Scenario 1: Simulate Normal Sinus Rhythm

1. HR = 70 bpm
2. All conductances = 1.0
3. All refractories = defaults
4. af_enabled = false
5. ectopic_enabled = false

6.1.2 Scenario 2: Simulate First-Degree AV Block

1. HR = 70 bpm
2. av_delay = 0.25 s (prolonged from normal 0.15 s)
3. av_conductance = 1.0 (1:1 conduction maintained)
4. Result: PR interval ≥ 200 ms

6.1.3 Scenario 3: Simulate Left Bundle Branch Block

1. HR = 70 bpm
2. lbb_conductance = 0.0 (complete block)
3. rbb_conductance = 1.0 (normal)
4. Result: Broad QRS (≥ 120 ms), wide S wave in V5–V6

6.1.4 Scenario 4: Simulate Atrial Fibrillation

1. HR = 130 bpm (controlled ventricular rate)
2. af_enabled = true
3. af_rate = 450 bpm (fibrillatory waves)
4. af_rr_distribution = poisson
5. Result: Irregular rhythm, absent P waves, f-waves visible

7 Real-Time Display Updates

7.1 Refresh Mechanism

- GUI updates at 30 FPS
- ECG trace accumulates every 33 ms
- Leads redrawn continuously
- Parameters can be adjusted without stopping simulation

7.2 Performance Notes

- Phase 1/2 engine: 1000 Hz update rate → GPU-limited by display
- Phase 3 engine: 150 Hz (2.2× real-time) → smooth real-time performance
- Parameter changes take effect immediately on next beat
- HR changes cause immediate cycle_length update